

Game Design Document (GDD): Squerre Escape

1. Short Description

Squerre Escape is a fast-paced 2D action platformer where players control a small blue square named Squerre trying to break free from captivity. Players must traverse dangerous terrain filled with floating platforms, spikes, patrolling foes, and automated turrets, all while under the watchful eye of a looming background boss named the Giant Watcher. The game blends high-mobility acrobatic platforming with a red light, green light survival mechanic.

2. Core Gameplay

The core loop centers around high-speed movement, precise aerial maneuvers, and survival reflexes. The player navigates obstacles using standard movement, double jumping, and high-velocity dashing. Mobility is enhanced by a grappling hook that latches onto rope blocks, turrets, and enemies, alongside a targeted dash attack system to strike down threats. Movement must be timed carefully: whenever the Giant Watcher turns forward, the player must come to a dead halt to avoid instant death.

3. Game Flow

The game begins at the Main Menu, which shows the best recorded escape time and a start prompt. Upon starting, the player spawns chained to a post and must dash to break the chain and trigger the escape timer. The player progresses horizontally across the stage toward the right, breaking obstacles, freeing an ally, and dodging or defeating enemies. If the player loses all health, lands on spikes, falls into the void, or moves while the Giant Watcher is watching, it results in a Game Over. Reaching and entering the Goal portal triggers a Win state, displaying the final run time and saving any new record.

4. Game Mechanics

Character Movement: Keys A and D control left and right movement, Space executes standard jumps and a mid-air double jump, and Shift or K triggers a quick dash. **Grappling Hook:** Left-clicking fires a grapple line toward rope blocks, turrets, or enemies. It allows swinging from fixed anchor points or pulling movable targets closer. **Target Lock and Dash Attack:** Holding right-click locks onto nearby enemies or turrets. Releasing right-click or pressing Shift launches a targeted dash attack straight at the locked entity, resetting the dash cooldown on impact. **Giant Watcher:** A giant cyclops-like overseer in the background. When its monitor is on and its back is turned, the player can move safely. When it turns to face forward, any movement or residual momentum immediately kills the player. **Interactive Obstacles:** Wooden barriers can be shattered with a dash. Metal barriers take multiple hits from projectiles and knock the player back while causing a dizzy status if dashed into. Spikes deal fatal damage upon contact. **Tri-Anger Ally:** A triangular ally trapped inside a cage that can be freed using a dash. Once liberated, it hovers above the player and fires bursts of projectiles at nearby threats. Players can toggle its firing behavior on or off using Q. **Enemies and Turrets:** Patrolling red circle enemies can be eliminated by stomping them or dashing through them. Stationary turrets aim and fire red

bullets toward the player at regular intervals. Built-in Level Editor: Pressing B toggles a full in-game editor to place blocks, enemies, traps, or pick-ups, test custom layouts, and export stage configurations.

5. Score and Progress System Progress Tracking

A persistent progress bar at the bottom of the screen tracks the player's horizontal position between the starting point and the escape portal, displaying completion from 0 to 100 percent. Time Score: Player performance is measured by completion time in seconds. The run timer begins the moment the starting chain is broken and stops upon entering the portal. If the completion time is lower than the previous record, it is stored as the new Best Time and displayed on both the victory screen and the main menu.